Типовик

02.10.23

Задание 1

а) $\sum\limits_{k=1}^{n} k(p^2-k^2) = \frac{1}{4}n(n+1)(2p^2-n^2-n)$; $p \in \mathbb{N}$

База $n=1$, тогда

$1(p^2-1) = \frac{1}{\cancel{2}} \cdot (\cancel{2}p^2 - \cancel{2})$; $(p^2-1) = (p^2-1)$ //

Предположим, что для m равенство тоже верно,

тогда $\sum\limits_{k=1}^{m} m(p^2-m^2) = \frac{1}{4}m(m+1)(2p^2-m^2-m)$

Переход $m \to m+1$

$$\sum_{k=1}^{m+1}(m+1)(p^2-(m+1)^2) = \frac{1}{4}(m+1)(m+2)(2p^2-(m+1)^2-(m+1))$$

$$\sum_{k=1}^{m+1}(m+1)(p^2-(m+1)^2) = \sum_{k=1}^{m} m(p-m^2) + (m+1)(p-(m+1)^2) \Rightarrow$$

$$\frac{1}{4}m\cancel{(m+1)}(2p^2-m^2-m)+\cancel{(m+1)}(p^2-(m+1)^2)=\frac{1}{4}\cancel{(m+1)}(m+2)$$
$$\cdot(2p^2-(m+1)^2-(m+1))$$

$$\underbrace{\frac{1}{4}m(2p^2-m^2-m)+(p^2-m^2-2m-1)}_{(*)}=\underbrace{\frac{1}{4}(m+2)(2p^2-m^2-2m-1-m-1)}_{(*)}$$

$(*)$: $\frac{1}{4}(m+2)(2p^2-m^2-3m-2)=\left(\frac{1}{4}m+\frac{1}{2}\right)(2p^2-m^2-3m-2)$

$$\cancel{\frac{1}{2}p^2m}-\cancel{\frac{1}{4}m^3}-\frac{3}{4}m^2-\frac{1}{2}m+\cancel{p^2}-\frac{1}{2}m^2-\frac{3}{2}m\cancel{-1}$$

$(*)$: $\cancel{\frac{1}{2}p^2m}-\cancel{\frac{1}{4}m^3}-\frac{1}{4}m^2+\cancel{p^2}-m^2-2m\cancel{-1}\Longrightarrow$

$$-\frac{3}{4}m^2-\frac{1}{2}m-\frac{1}{2}m^2-\frac{3}{2}m=-\frac{1}{4}m^2-m^2-2m \quad \Big|\cdot(-4)$$

$$3m^2+2m+2m^2+6m=m^2+4m^2+8m$$

$$5m^2+8m=5m^2+8m$$

$$\#$$

б) $2(\sqrt{n+1}-1)<\frac{1}{\sqrt{1}}+\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}+\ldots+\frac{1}{\sqrt{n}}$

База $n=1$ $\quad 0<1$ #

Предположим, что для m неравенство, тоже верно, $2(\sqrt{m+1}-1)<\frac{1}{\sqrt{1}}+\frac{1}{\sqrt{2}}+\dots\frac{1}{\sqrt{m}}$

Переход $m\to m+1$: $2(\sqrt{m+2}-1)<\frac{1}{\sqrt{1}}+\frac{1}{\sqrt{2}}+\dots+\frac{1}{\sqrt{m}}+\frac{1}{\sqrt{m+1}}$

$$2(\sqrt{m+2}-1)<2(\sqrt{m+1}-1)+\frac{1}{\sqrt{m+1}}$$

$$2\sqrt{m+2}-\cancel{2}<2\sqrt{m+1}-\cancel{2}+\frac{1}{\sqrt{m+1}}$$

$$2\sqrt{m+2}<\frac{2m+3}{\sqrt{m+1}}$$

$$4m+8<\frac{4m^2+12m+9}{m+1}\ ;\ m.к\ m\in\mathbb{N}\to$$

$$4\cancel{m^2}+\cancel{12m}+8<\cancel{4m^2}+\cancel{12m}+9$$

$$8<9$$

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Задание 2.

а) $C_n^0-2C_n^1+3C_n^2-\dots+(-1)^n(n+1)C_n^n=C_n^n-2C_n^{n-1}+3C_n^{n-2}-\dots+(-1)^n(n+1)C_n^0=A$, тогда

$2A=(C_n^0-2C_n^{-1}+3C_n^2-\dots+(-1)^n(n+1)C_n^n)$

$$2A = (C_n^n - 2C_n^{n-1} + 3C_n^{n-2} - \dots + (-1)^n (n+1) C_n^0)$$

$$+ C_n^n - 2C_n^{n-1} + 3C_n^{n-2} - \dots + (-1)^n (n+1) C_n^0$$

$$2A = C_n^0 (1-1)^n$$